

A5

Equal-area cells for geographic data.

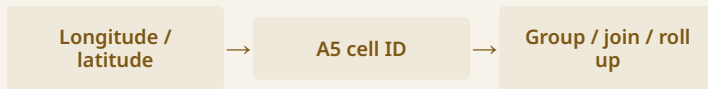


Install & load

```
INSTALL a5 FROM community;
LOAD a5;
```

01 At a glance

Turn longitude and latitude into pentagonal cell IDs for grouping, joining, and geographic rollups. A5 uses equal-area cells at each resolution, with integer IDs and functions for hierarchy, neighborhoods, and geometry conversion.



02 Core SQL surface

`a5_lonlat_to_cell(lon,lat,res)`

Map longitude, latitude in degrees to a UBIGINT cell ID.

`a5_cell_to_lonlat(cell)`

Return the cell center as [longitude, latitude], not the original point.

`a5_cell_to_parent(cell,res)`

Find a coarser ancestor for hierarchical rollups.

`a5_cell_area(res)`

Return equal cell area in square meters at a resolution.

`a5_grid_disk(cell,k)`

Return cells within k edge steps, including the center cell.

`a5_cell_to_geometry(cell)`

Produce a native GEOMETRY polygon for mapping.

`a5_geometry_to_cells(geom,res)`

Cover geometry with a compacted list of cell IDs.

03 Keep in mind

Coordinates and identifiers

Pass longitude first, latitude second, in degrees. Resolutions run from 0 to 30; larger values mean smaller cells. Preserve IDs as UBIGINT or use `a5_u64_to_hex` when exporting to systems with limited integer precision.

04 Encode New York City

```
SELECT a5_lonlat_to_cell(
  -74.0060, 40.7128, 15
) AS cell;
SELECT a5_cell_area(15) AS area_m2;
```

Cell 2742821848331845632; area ≈31,669 m².

05 Count observations by cell

```
WITH points(lon,lat) AS (VALUES
  (-74.0060,40.7128), (-74.0061,40.7129),
  (-73.9857,40.7484)
)
SELECT a5_lonlat_to_cell(lon,lat,15) AS cell,
       count(*) AS n
FROM points
GROUP BY cell;
```

Equal-resolution cell IDs are ordinary grouping and join keys.

06 Roll up and draw a cell

```
WITH p AS (
  SELECT a5_lonlat_to_cell(
    -74.0060,40.7128,15) AS cell
)
SELECT a5_cell_to_parent(cell,10) AS parent,
       a5_cell_to_geometry(cell) AS polygon
FROM p;
```

The parent is a hierarchy key; the polygon is native GEOMETRY.

07 In practice

Render at one resolution

`a5_geometry_to_cells` returns compacted coverage. Uncompact with `a5_uncompact(cells, res)` before drawing cell polygons together. Native GEOMETRY output works without spatial; load spatial only when you need its additional geometry functions.

[Full docs & examples ↗](#) [Source ↗](#)